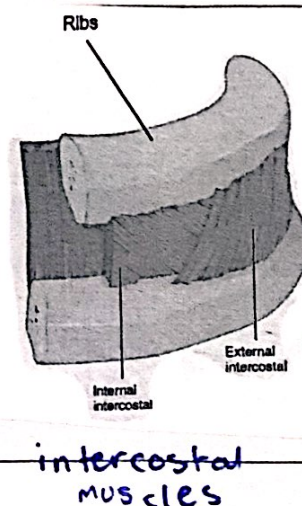
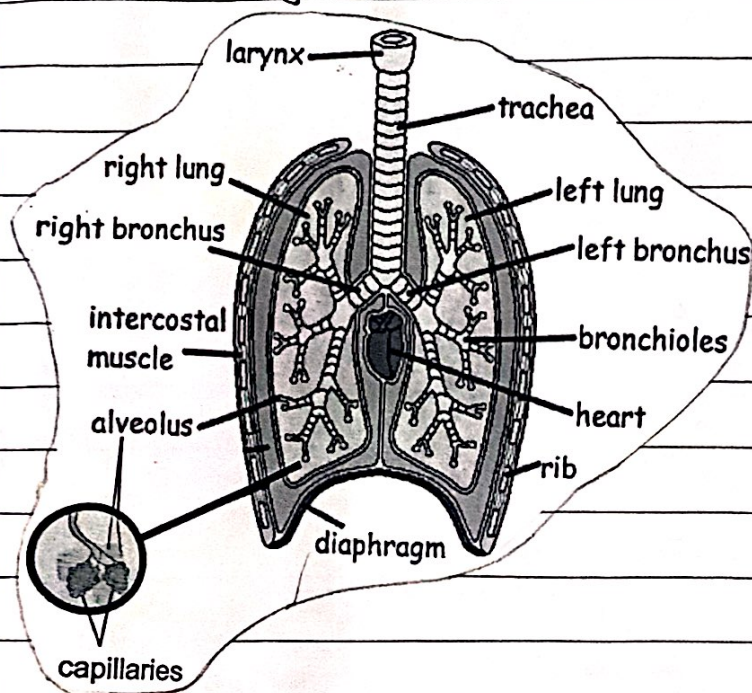


Features of gas exchange surfaces::

- Large surface area to allow faster diffusion
- thin walls to shorten diffusion distances
- good ventilation with air to maintain diffusion gradient
- good blood supply to maintain a high concentration gradient for faster diffusion

The breathing system::



- the cartilage in the trachea provides support and keeps the trachea from collapsing

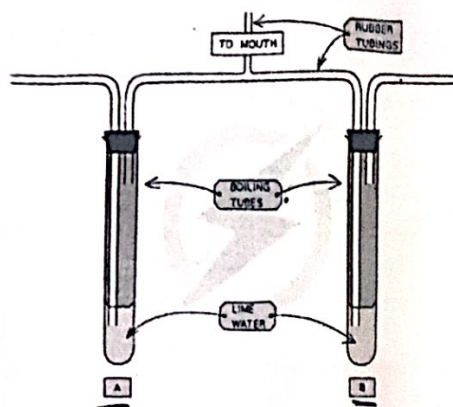
* ribs protect internal organs. Intercostal muscles control the ribs movement causing inhalation/exhalation. Diaphragm is a sheet of connective tissue & muscle that helps change the volume of the thorax (area between neck and abdomen) to allow the inhalation & exhalation,

Inspired air

- 21% oxygen
- 0.04% CO_2
- 78% Nitrogen
- 0.5 water vapor

Expired air

- 16% oxygen
- 4% CO_2
- 78% Nitrogen
- 1% water vapor



- * breathe in = drawn from tube A
- * breathe out = blown into tube B
- limewater in tube A will remain clear but the lime water in tube B will be cloudy

% of CO_2 (exhaled) > % of CO_2 (inhaled)

Physical Activity

increases the frequency / depth of breathing
the breaths taken & average chest expansion
will be higher after exercise than before.

↳ because there is an increase in CO_2
concentration, detected by the brain, leading
to an increased rate and greater depth
of breathing,

Goblet cells, mucus & ciliated cells :-

- mucus is made by special mucus-producing cells called goblet-cells.
- the mucus traps bacteria, dust, pathogens etc.
- * the cilia beats and pushes the mucus away from the lungs and towards the throat